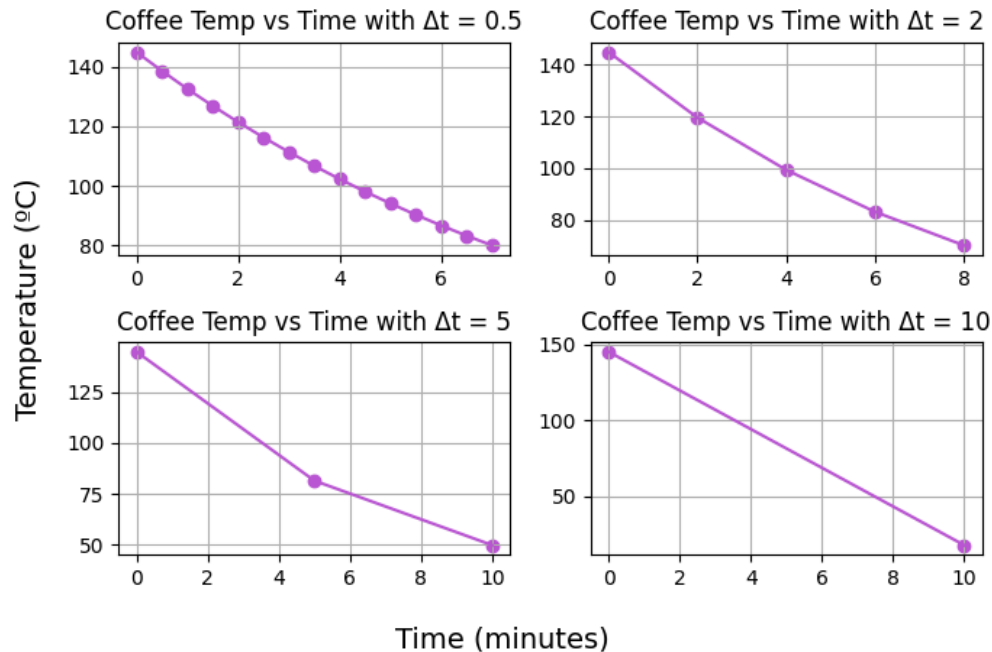


Maddy St. Charles
Climate 410, HW3
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Coffee Cooling over Time with Different Time Steps



The figure above depicts the temperature of the coffee as time progresses, each subplot with a different Δt . I created a function that takes the initial temperature of the coffee, the drinkable temperature, the temperature of its environment, and the desired Δt and uses Euler's method to estimate the time it takes for the coffee to reach the drinkable temperature. Throughout this process, the temperature of the coffee is calculated at each increment of Δt , which is what is plotted above for four values of Δt . It is evident that a smaller value of Δt provides a more accurate picture of how the coffee's temperature changes with time. For example, when $\Delta t=10$, there are only two data points, creating a straight line straight from the original temperature of 145°F to the drinkable temperature of 80°F. However, the plot of $\Delta t=5$ shows a bit of curvature to the data, even though it only has three data points. Decreasing Δt down to 0.5 makes the true shape of the plot even clearer. It is not a perfectly straight line, but rather has a bit of a curve to it. Based on these results, it can be concluded that a greater Δt is generally more beneficial when estimating the behavior of a set of variables.